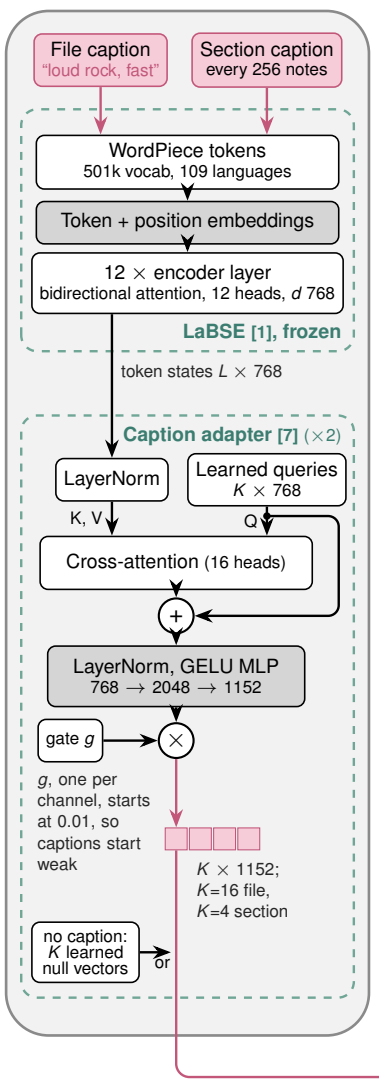
Captioned music model (477M)

477M trainable parameters: trunk 446M, cascade heads 16M, two caption adapters 13M, embeddings 2M. The LaBSE text encoder (471M) is frozen and not counted.

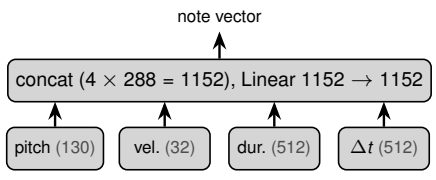
Status: the music model without caption adapters (464M) is trained. The caption stage of this layout (F1) is planned; caption results here come from 107M pilots.
F1 will also add a length-control input and infilling (roadmap 11–12); they are not drawn here yet.

How to read: numbers give the reading order; pink = captions; grey boxes = linear layers and embedding tables; dashed frame = a module drawn open in place; dotted frame or box = a zoom (white panel) into the dotted part.

1 Caption path



2 Input embedding

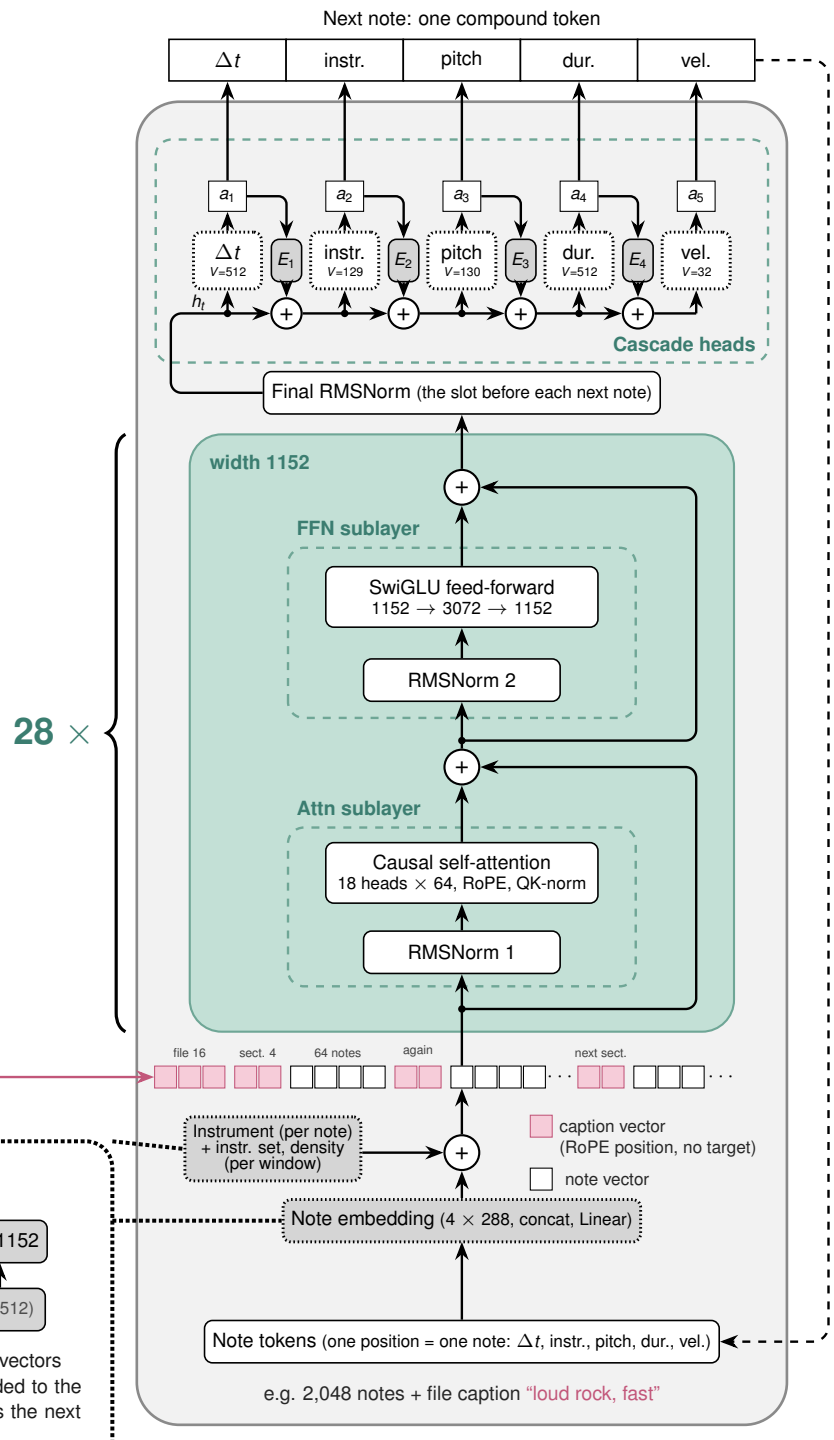


+ instrument of this note: a table of 129 vectors (128 General MIDI programs + drums), added to the note vector. The output head *instr.* predicts the next one.

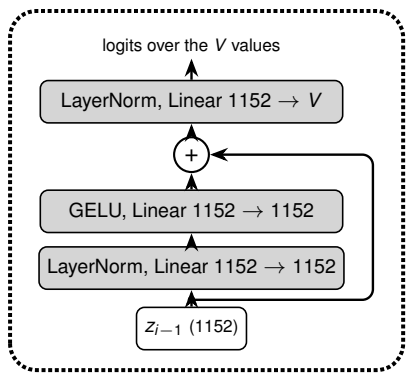
+ window conditions, the same vector at every note position: the **instrument set** of the window (129-dim multi-hot → Linear) and its **density** (notes per second in 15 levels + "none" → a table of 16 vectors). Both start at zero and are dropped 15% of the time in training, so "none" stays usable. In generation they come from the user or are measured on the prompt.

Time is absolute: Δt and durations in 20 ms steps, clipped at 10.22 s. There are no bars, beats or tempo tokens; the metre has to be read from the timing. 2,048 notes are about 1–4 minutes of music.

3 Music model



4 One cascade head (the 5 dotted boxes, own weights)



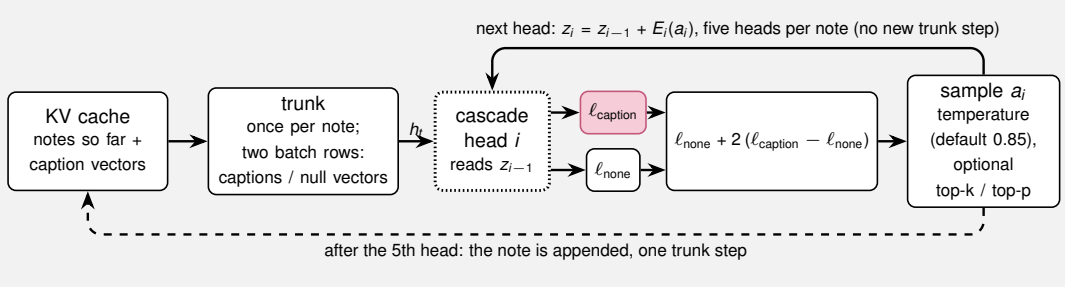
Why a cascade. One position predicts all five values of the next note. Head i reads the running state

$$z_0 = h_t, \quad z_i = z_{i-1} + E_i(a_i),$$

the trunk state h_t plus the embeddings E of the values already chosen. So the note's probability follows the chain rule, $p(\Delta t)\, p(\text{instr.} \mid \Delta t)\, p(\text{pitch} \mid \Delta t, \text{instr.}) \cdots$ (compound tokens as in [6]).

a_i is the true value in training (teacher forcing) and the sampled value in generation. Loss: the sum of the five cross-entropies.

5 Generation



Classifier-free guidance [9] with scale 2: every step runs the batch twice as wide, once with the captions and once with null vectors.

Caption vectors enter the cache as in training: the file caption first, then a section's 4 vectors at its first note and again every 64 notes. Past 2,048 notes the window slides by a quarter.

6 Training recipe for section captions

Without these measures the model follows the first section's caption and then ignores later ones.

1. Change-aware captions. Each section caption names what changed since the section before ("drums enter, louder"). Half of the training windows contain a section change.

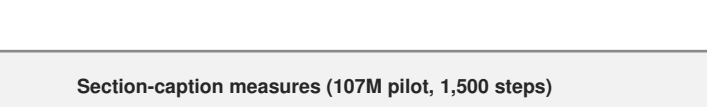
2. Spliced windows (25% of windows).

File A up to a section boundary, then a section of file B (dashed) with B's caption and no file caption: the caption contradicts the context and has to win.

3. Caption repeated every 64 notes, in training and in generation.

The same 4 section vectors are inserted again, so the caption is never far back in the context.

4. Counterfactual loss [8] + recognition head.



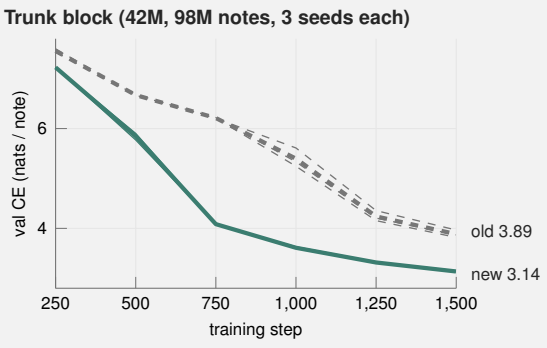
NLL is measured on the first 32 notes of a changed section. The true caption must beat the previous one by 0.05 nats (weight 1.0). A linear head on those notes' hidden states tells the true caption from the previous one (BCE, weight 0.1). Both are used in training only.

5. Caption dropout → guidance. File captions are replaced by null vectors 15% of the time, section captions 20%, so generation can use classifier-free guidance [9] with scale 2.

Evidence so far (107M pilot, 1,500 steps, one seed). Switch test: 6 prompt pairs A/B (e.g. soft / loud), 16 samples each, the section caption changes at note 1,024, guidance 2. The matching attribute (velocity, notes per second, register, drum or string share) is measured per 256 notes; the score is its change after the switch divided by its offset before it (0 = keeps the first style, 1 = a full flip; single prompts reach 1.5). The start model v2_A0: 0.01. 1,500 more steps without measures 2–4 (control): **0.06**, held-out NLL 2.772; with all three: **0.85**, NLL 2.783; splice alone 0.59, counterfactual alone 0.13, repeat alone 0.09. Open cost: holding a caption overshoots (strings 15 notes per onset, 44% of samples run away; the fix is being tested). No error bars yet.

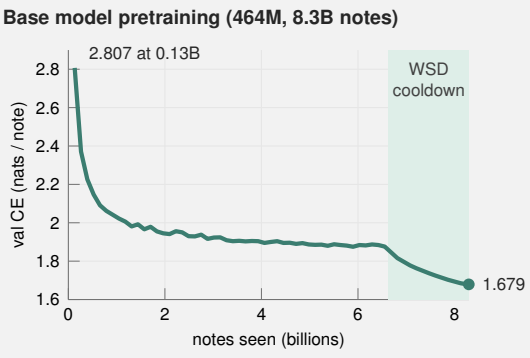
7 Why these choices: training curves (teal = what the model uses)

Trunk block (42M, 98M notes, 3 seeds each)



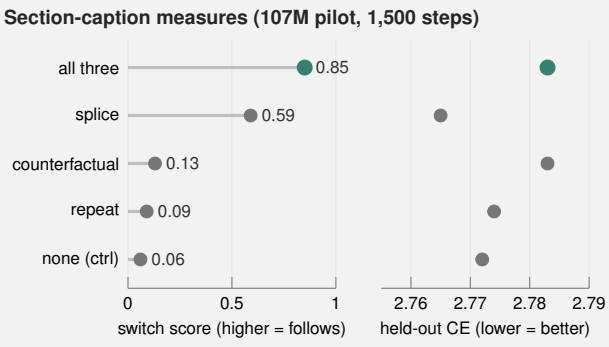
Clean GigaMIDI validation. **New**: RMSNorm, SwiGLU, QK-norm; **old** (dashed): LayerNorm, GELU; both RoPE and Muon. Thin lines = seeds, thick = mean; −0.75 at the end.

Base model pretraining (464M, 8.3B notes)



Clean GigaMIDI validation every 2,000 steps (65,536 notes per step); constant LR, then the $1 - \sqrt{\cdot}$ cooldown in the last 20%. One run on one RTX 5090, ≈41 h.

Section-caption measures (107M pilot, 1,500 steps)

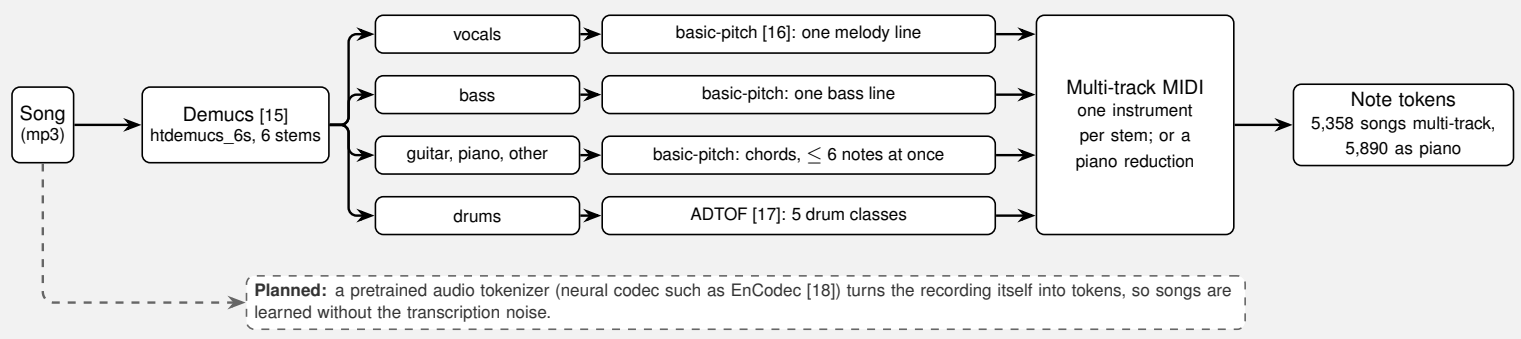


Each row continues the same model (v2_A0) for 1,500 steps with one measure, or all three. Switch score as in section 6; one seed, no error bars. **Teal** = the recipe in use.

8 Training stages

Setting	Base pretraining (done)	Caption training, F1 run (planned)	Fine-tune on Ukrainian songs
Start	from scratch	the base + the two new caption adapters; also planned, not drawn: a length-control input and infilling (roadmap 11–12)	the base (done, no captions); the F1 model with captions (F2, planned)
Data	GigaMIDI [12] 0.4, Aria-MIDI [13] 0.3, Discover MIDI [19] 0.3; 8.3B notes (1.8 epochs on average; Aria ≈4.5)	captioned MIDI in 3 tiers: MidiCaps [14] / XMIDI [20] / GigaMIDI metadata, Aria-MIDI metadata, the rest with feature captions; 2.5–4B notes	5,358 songs transcribed from audio (section 9; 4,811 train, 547 val) 0.75 + GigaMIDI replay 0.25 against forgetting
Captions	none: only the instrument-set and density conditions	templates filled from metadata (genre, mood, artist, composer) and from features computed from the notes (tempo, loudness, register, instruments), 30% in Ukrainian; MidiCaps' LLM-written captions; section captions from each section's own notes	none in the done run; F2 adds them
Window	2,048 notes (≈1–4 min)	2,048 notes + 16 file vectors + 4 per caption arrival (section start and every 64 notes): 2,212 positions (37 caption slots per window, fixed shape; ≈2,192 used)	2,048 notes
Batch	65,536 notes per step (32 windows)	same	24,576 notes per step (12 windows)
Optimizer, LR	Muon [10] for weight matrices, AdamW for the rest; one LR 10^{-3} (Muon updates scaled to AdamW's size)	AdamW, LR 10^{-4}, adapters 10× (as in the 107M pilots)	8-bit AdamW, LR $2 \cdot 10^{-4}$
Schedule	WSD [11]: warmup, stable (constant LR), $1 - \sqrt{\cdot}$ decay over the last 20%	same; planned: section captions and the counterfactual loss weighted up in the decay (roadmap 13)	same
Steps	126,582 (≈41 h on one RTX 5090)	≈38–61k	1,500
Precision	bf16 with fp8 matmuls, torch.compile	same	same

9 Audio → MIDI: how songs become training notes (fine-tune data)



Every step adds errors: stem bleed, missed or extra notes, octave and timing errors, drum classes mixed up, loudness guessed from activations; stems quieter than −30 dB vs the mix are dropped. The model learns these errors as if they were music.

10 Glossary

Δt	time since the previous note started (20 ms steps, 512 values)	**RoPE [2]**	rotary position embedding: positions as rotations of queries and keys
instr.	instrument: General MIDI program 0–127 or drums (129 values)	**RMSNorm [3]**	layer normalisation by the root mean square only
pitch	MIDI pitch 0–127 + start / end of piece (130 values)	**SwiGLU [4]**	feed-forward layer with a SiLU-gated linear unit
dur.	note length (20 ms steps, 512 values)	**QK-norm [5]**	queries and keys normalised before the dot product
vel.	velocity: how hard the note is played (32 levels)	**LaBSE [1]**	multilingual sentence encoder (109 languages)
compound token	one sequence position = one whole note	**NLL**	negative log-likelihood of the true notes
caption vector	a caption-adapter output placed in the sequence; no target of its own, but the last vector of a section caption predicts the next note	**nats**	units of NLL with the natural log
null vectors	learned stand-ins for "no caption"	**BCE**	binary cross-entropy (two-class loss)
teacher forcing	training feeds the true earlier values, not samples	**CFG [9]**	classifier-free guidance: push logits away from the caption-free prediction
KV cache	keys and values of earlier positions kept between steps	**WSD [11]**	warmup, stable, decay: LR schedule with a short final cooldown
multi-hot	a 0/1 vector with a 1 for every instrument present	**Muon [10]**	optimizer that orthogonalises the update of each weight matrix
stem	one separated part of a recording (vocals, bass, drums, ...)	**Demucs [15]**	neural source separation: a song into 6 stems

11 References (links)

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[6] Compound Word Transformer (2021), arXiv:2101.02402
[7] Flamingo: learned-query resampler, gated adapters (2022), arXiv:2204.14198
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[9] Classifier-Free Diffusion Guidance (2022), arXiv:2207.12598
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[14] MidiCaps: MIDI with text captions (2024), arXiv:2406.02255
[15] Demucs: Hybrid Transformers for Music Source Separation (2022), arXiv:2211.08553
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[19] Discover MIDI Dataset, Hugging Face dataset
[20] XMusic: the XMIDI dataset (2025), arXiv:2501.08809